

Antenna Digital Twin: Formulas and Calculations Reference

This document lists the formulas used in the Antenna Digital Twin codebase and the purpose of each calculation.

1 Electromagnetic Solver (Patch Antenna)

1.1 Effective Permittivity

Used to account for fringing fields in a rectangular patch (Hammerstad-style).

$$\varepsilon_{\text{eff}} = \frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{2} \left(1 + \frac{12h}{W}\right)^{-1/2} \quad (1)$$

Used for: Resonance frequency and quality factor.

Location: `backend/em_solver/adapters/meep_adapter.py`

1.2 Effective Length (Fringing Extension)

Extension of patch length due to fringing (effective length correction).

$$\Delta L = 0.412 h \frac{\varepsilon_{\text{eff}} + 0.3}{\varepsilon_{\text{eff}} - 0.258} \frac{W/h + 0.264}{W/h + 0.8} \quad (2)$$

$$L_{\text{eff}} = L + 2\Delta L \quad (3)$$

Used for: Resonance frequency f_{res} .

Location: `backend/em_solver/adapters/meep_adapter.py`

1.3 Resonance Frequency

Cavity-model resonance for a rectangular patch.

$$f_{\text{res}} = \frac{c_0}{2 L_{\text{eff}} \sqrt{\varepsilon_{\text{eff}}}} \quad (4)$$

with $c_0 = 299792458$ m/s.

Used for: Center frequency of S11 response and normalized frequency variable.

Location: `backend/em_solver/adapters/meep_adapter.py`

1.4 Quality Factor (Patch Q)

Thin-patch cavity / Wheeler-style Q for bandwidth.

$$Q_{\text{patch}} = \frac{\lambda_{\text{res}} \sqrt{\varepsilon_{\text{eff}}}}{4 h \sqrt{\varepsilon_r}} \quad (5)$$

with $\lambda_{\text{res}} = c_0/f_{\text{res}}$. Clipped to $[10, 80]$ in code.

Used for: S11 magnitude and phase shape; fractional bandwidth $\approx 0.667/Q$.

Location: `backend/em_solver/adapters/meep_adapter.py`

1.5 Normalized Frequency

$$x = \frac{f - f_{\text{res}}}{f_{\text{res}}} \quad (6)$$

Used for: Resonant S11 model (RLC-like response).

1.6 S11 Magnitude (Resonant Model)

Single-resonance RLC-like model for $|S_{11}|^2$:

$$|S_{11}|^2 = \frac{(Q_{\text{patch}} x)^2}{1 + (Q_{\text{patch}} x)^2} \quad (7)$$

$$|S_{11}| = \sqrt{\max(0, \min(1, |S_{11}|^2))} \quad (8)$$

Used for: S11 magnitude vs frequency when using geometry-based approximation.

Location: backend/em_solver/adapters/meep_adapter.py

1.7 S11 Phase

$$\angle S_{11} = -\frac{180}{\pi} \arctan(Q_{\text{patch}} x, 1) \quad (9)$$

(Phase in degrees; `arctan2` convention.)

Used for: Complex S_{11} and input impedance.

1.8 S11 in Decibels

$$S_{11,\text{dB}} = 20 \log_{10}(\max(|S_{11}|, 10^{-9})), \quad S_{11,\text{dB}} \leq 0 \text{ dB} \quad (10)$$

Used for: S11 magnitude in dB for plots and metrics (e.g. S11 min, bandwidth).

1.9 Input Impedance from S11

Reference impedance $Z_0 = 50 \Omega$:

$$Z_{\text{in}} = Z_0 \frac{1 + S_{11}}{1 - S_{11}} \quad (11)$$

Used for: Logging and consistency; S_{11} is complex here.

Location: backend/em_solver/adapters/meep_adapter.py

1.10 Aperture Directivity

Physical optics-style aperture directivity for a rectangular patch:

$$D_{\text{aperture}} = \frac{4\pi L W}{\lambda_0^2} \quad (12)$$

with $\lambda_0 = c_0/f_0$.

Used for: Base directivity before geometry factor.

Location: backend/em_solver/adapters/meep_adapter.py

1.11 Directivity (with Geometry Factor)

$$D_{\text{linear}} = \max(D_{\text{aperture}} (2 + \min(W/L, 1)), 1) \quad (13)$$

Used for: Linear directivity before efficiency.

Location: backend/em_solver/adapters/meep_adapter.py

1.12 Power Transmitted (from S11)

$$P_{\text{transmitted}} = 1 - 10^{S_{11,\text{min,dB}}/10} \quad (14)$$

Used for: Fraction of power accepted by the antenna (mismatch).

Location: backend/em_solver/adapters/meep_adapter.py

1.13 Substrate Loss (Dielectric)

Exponential attenuation in substrate:

$$\eta_{\text{substrate}} = 1 - \exp\left(-\frac{2\pi f_0 \tan \delta h \sqrt{\epsilon_r}}{c_0}\right) \quad (15)$$

Used for: Efficiency reduction due to dielectric loss.

Location: backend/em_solver/adapters/meep_adapter.py

1.14 Radiation Efficiency

$$\eta = P_{\text{transmitted}} (1 - \eta_{\text{substrate}}) \quad (16)$$

Optionally reduced by feed-offset factor; then clipped to $[0, 1]$.

Used for: Efficiency field in EMSimulationResult.

Location: backend/em_solver/adapters/meep_adapter.py

1.15 Gain (dBi)

$$G_{\text{linear}} = D_{\text{linear}} \eta, \quad G_{\text{dBi}} = 10 \log_{10}(\max(G_{\text{linear}}, 10^{-6})) \quad (17)$$

Used for: Peak gain in dBi in simulation results.

Location: backend/em_solver/adapters/meep_adapter.py

1.16 Substrate Conductivity (OpenEMS)

Equivalent conductivity for lossy substrate in FDTD:

$$\kappa = \tan \delta \cdot 2\pi f_0 \epsilon_0 \epsilon_r \quad (18)$$

Used for: Kappa of substrate material in OpenEMS.

Location: backend/em_solver/adapters/openems_adapter.py

1.17 Mesh Resolution (OpenEMS)

$$\Delta_{\text{mesh}} = \frac{c_0}{f_0 \cdot 20} \quad (19)$$

(20 cells per wavelength at f_0 .)

Used for: Grid spacing in OpenEMS FDTD.

Location: backend/em_solver/adapters/openems_adapter.py

2 Propagation and Coverage

2.1 Free-Space Path Loss

$$L_{\text{path}} = 20 \log_{10}\left(\frac{4\pi d}{\lambda}\right) \quad (20)$$

with $\lambda = c/f$, d in m, f in Hz.

Used for: Path loss in dB for coverage and link budget.

Location: backend/environment/propagation.py

2.2 Indoor / Outdoor Offsets

Indoor: $L_{\text{path}} + 20$ dB; outdoor: $L_{\text{path}} + 10$ dB.

Used for: Simple environment-dependent path loss.

Location: backend/environment/propagation.py

2.3 Effective Gain and Coverage

$$G_{\text{eff}} = G_{\text{max}} - L_{\text{path}} \quad (21)$$

Coverage: probability 1 if $G_{\text{eff}} > G_{\text{threshold}}$, else 0.

Used for: Coverage probability and effective gain.

Location: backend/environment/coverage.py

3 ROM, Sensitivity, and Error Bounds

3.1 Parameter and Output Normalization

Z-score normalization:

$$x_{\text{norm}} = \frac{x - \bar{x}}{\sigma_x + 10^{-10}}, \quad y_{\text{norm}} = \frac{y - \bar{y}}{\sigma_y + 10^{-10}} \quad (22)$$

Used for: Sensitivity analysis (Sobol proxy) and scaling.

Location: backend/rom/sensitivity_analysis.py

3.2 Sobol Indices (Correlation-Based Proxy)

First-order sensitivity proxy using Pearson correlation:

$$S_i \propto |\rho(x_{\text{norm},i}, y_{\text{norm}})| \quad (23)$$

Normalized so that $\sum_i S_i = 1$.

Used for: Parameter ranking for ROM construction.

Location: backend/rom/sensitivity_analysis.py

3.3 Prediction Intervals (Error-Based)

Errors $e = y_{\text{true}} - y_{\text{pred}}$; interval using mean and std of e and normal quantile z :

$$[y_{\text{pred}} - \bar{e} - z\sigma_e, y_{\text{pred}} - \bar{e} + z\sigma_e] \quad (24)$$

with $z = \Phi^{-1}((1 + \text{confidence})/2)$.

Used for: Confidence bounds on ROM predictions.

Location: backend/rom/error_bounds.py

3.4 Per-Metric Error Statistics

$$\text{MAE} = \frac{1}{n} \sum |e_i| \quad (25)$$

$$\text{RelErr} = \frac{|e_i|}{|y_{\text{true},i}| + 10^{-10}} \quad (26)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum e_i^2} \quad (27)$$

$$\text{MaxError} = \max_i |e_i| \quad (28)$$

Percentiles (e.g. 95th, 99th) of $|e_i|$ are also computed.

Used for: Error bounds per metric (s11_min, gain, efficiency).

Location: backend/rom/error_bounds.py

3.5 Uncertainty Growth with Distance

Exponential growth of uncertainty with distance from training data:

$$u(d) = u_0 (1 + 0.1 d) \quad (29)$$

Used for: Uncertainty quantification when extrapolating.

Location: backend/rom/error_bounds.py

4 ML Training and Evaluation

4.1 MSE, MAE, RMSE, R²

$$\text{MSE} = \frac{1}{n} \sum (y_i - \hat{y}_i)^2, \quad \text{MAE} = \frac{1}{n} \sum |y_i - \hat{y}_i|, \quad \text{RMSE} = \sqrt{\text{MSE}} \quad (30)$$

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \quad (31)$$

Used for: Model evaluation (s11_min, gain, efficiency).

Location: backend/ml_models/training_pipeline.py

4.2 Gaussian Process (Surrogate)

Mean: constant; kernel: scaled Matérn ($\nu = 2.5$).

Used for: Surrogate model with uncertainty (mean and variance per output).

Location: backend/ml_models/gaussian_process.py

5 Summary Table

Quantity	Formula (summary)	Purpose
ε_{eff}	Eq. (1)	Fringing; f_{res} , Q
ΔL , L_{eff}	Eq. (2)–(3)	Resonance length
f_{res}	Eq. (4)	S11 center frequency
Q_{patch}	Eq. (5)	S11 bandwidth and shape
$ S_{11} $, $S_{11,\text{dB}}$	Eq. (7)–(9)	Return loss
Z_{in}	Eq. (10)	Input impedance
D , η , G_{dBi}	Eq. (11)–(15)	Gain and efficiency
L_{path}	Eq. (16)	Path loss (propagation)
G_{eff}	Eq. (18)	Coverage / link budget
Sobol / MAE / RMSE / R ²	§3–4	Sensitivity and ROM error